

10-Mark Questions & Answers — Data Structures Module 6: Sorting & Searching Techniques

Q1. Explain all major Sorting and Searching techniques, Hashing, and Collision Resolution methods in detail with complexity tables and C code.

Comprehensive 10-mark master guide covering:

1. Linear Search and Binary Search algorithms.
2. Bubble, Insertion, Selection, Merge, and Quick Sort algorithms.
3. Hash functions (Division, Mid-square, Folding, Multiplication).
4. Collision resolution: Open addressing (Linear, Quadratic, Double Hashing) vs Separate Chaining.
5. Complete Big-O Space and Time complexity comparison summary table.